



1.4 PRACTICAL EXAMPLES WITH RESISTORS

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Figure 1.5 shows two practical examples with nonlinear and regular resistors as trimmer potentiometers, elements which will be covered in the following chapter.

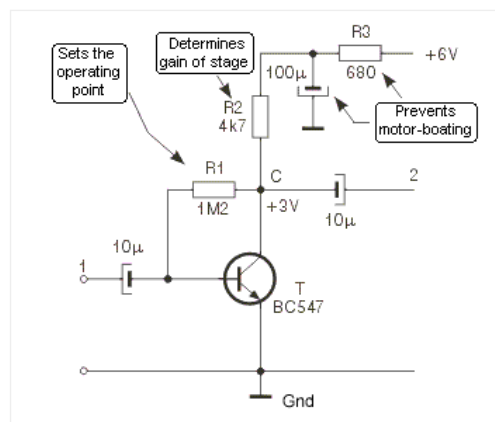


Fig. 1.5a: RC amplifier

Figure 1.5a represents an RC voltage amplifier, that can be used for amplifying low-frequency, low-amplitude audio signals, such as microphone signals. The signal to be amplified is brought between node 1 (amplifier input) and gnd, while the resulting amplified signal appears between node 2 (amplifier output) and gnd. To get the optimal performance (high amplification, low distortion, low noise, etc) , it is necessary to “set” the transistor’s operating point. Details on the operating point will be provided in chapter 4; for now, let’s just say that DC voltage between node C and gnd should be approximately one half of battery (power supply) voltage. Since battery voltage equals 6V, voltage in node C should be set to 3V. Adjustments are made via resistor R1.

Connect a voltmeter between node C and gnd. If voltage exceeds 3V, replace the resistor R1=1.2MW with a smaller resistor, say R1=1MW. If voltage still exceeds 3V, keep lowering the resistance until it reaches approximately 3V. If the voltage at node C is originally lower than 3V, increase the resistance of R1.

The degree of amplification of the stage depends on R2 resistance: *higher resistance – higher amplification, lower resistance – lower amplification*. If the value of R2 is changed, the voltage at node C should be checked and adjusted (via R1).

Resistor R3 and 100μF capacitor form a filter to prevent feedback from occurring. This feedback is called “Motor-boating” as it sounds like the noise from a motor-boat. This noise is only produced when more than one stage is employed.

As more stages are added to a circuit, the chance of feedback, in the form of instability or motor-boating, will occur.

This noise appears at the output of the amplifier, even when no signal is being delivered to the amplifier.

The instability is produced in the following manner:

Even though no signal is being delivered to the input, the output stage produces a very small background noise called “hiss. This comes from current flowing through the transistors and other components.

This puts a very small waveform on the power rails. This waveform is passed to the input of the first transistor and thus we have produced a loop for “noise-generation.” The speed with which the signal can pass around the circuit determines the frequency of the instability. By adding a resistor and electrolytic to each stage, a low-frequency filter is produced and this “kills” or reduces the amplitude of the offending signal. The value of R3 can be increased if needed.

Practical examples with resistors will be covered in the following chapters as almost all circuits require resistors.

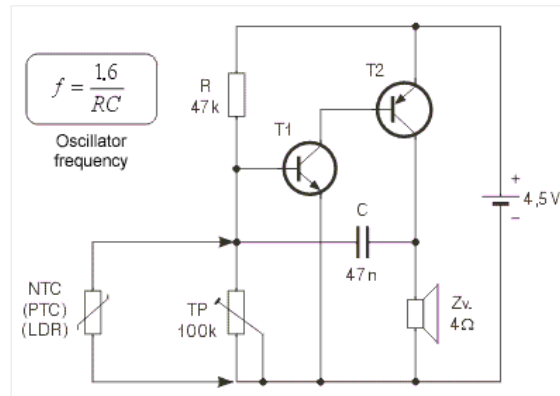


Fig. 1.5b: Sound indicator of changes in temperature or the amount of light

A practical use for nonlinear resistors is illustrated on a simple alarm device shown in figure 1.5b. Without trimmer TP and nonlinear NTC resistor it is an audio oscillator. Frequency of the sound can be calculated according to the following formula:

$$f = \frac{1.6}{RC}$$

In our case, $R=47\text{k}\Omega$ and $C=47\text{nF}$, and the frequency equals:

$$f = \frac{1.6}{47 \cdot 10^3 \cdot 47 \cdot 10^{-9}} \approx 724 \text{ Hz}$$

When, according to the figure, trim pot and NTC resistor are added, oscillator frequency increases. If the trim pot is set to minimum resistance, the oscillator stops. At the desired temperature, the resistance of the trim pot should be increased until the oscillator starts working again. For example, if these settings were made at 2°C , the oscillator remains frozen at higher temperatures, as the NTC resistor's resistance is lower than nominal. If the temperature falls the resistance increases and at 2°C the oscillator is activated.

If an NTC resistor is installed in a car, close to the road surface, the oscillator can warn driver if the road is covered with ice. Naturally, the resistor and two copper wires connecting it to the circuit should be protected from dirt and water.

If, instead of an NTC resistor, a PTC resistor is used, the oscillator will be activated when the temperature rises above a certain designated value. For example, a PTC resistor could be used for indicating the state of a refrigerator: set the oscillator to work at temperatures above 6°C via trimmer TP, and the circuit will signal if anything is wrong with the fridge.

Instead of an NTC, we could use an LDR resistor – the oscillator would be blocked as long as a certain amount of light is present. In this way, we could make a simple alarm system for rooms where a light must be always on.

The LDR can be coupled with resistor R. In that case, the oscillator works when the light is present, otherwise it is blocked. This could be an interesting alarm clock for huntsmen and fishermen who would like to get up at the crack of dawn, but only if the weather is clear. For the desired moment in the early morning, the trim pot should be set to the uppermost position. Then, the resistance should be carefully reduced, until the oscillator starts. During

the night the oscillator will be blocked, since there is no light and LDR resistance is very high. As the amount of light increases in the morning, the resistance of the LDR drops and the oscillator is activated when the LDR is illuminated with the required amount of light.

The trim pot from the figure 1.5b is used for fine adjustments. Thus, TP from figure 1.5b can be used for setting the oscillator to activate under different conditions (higher or lower temperature or amount of light).



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